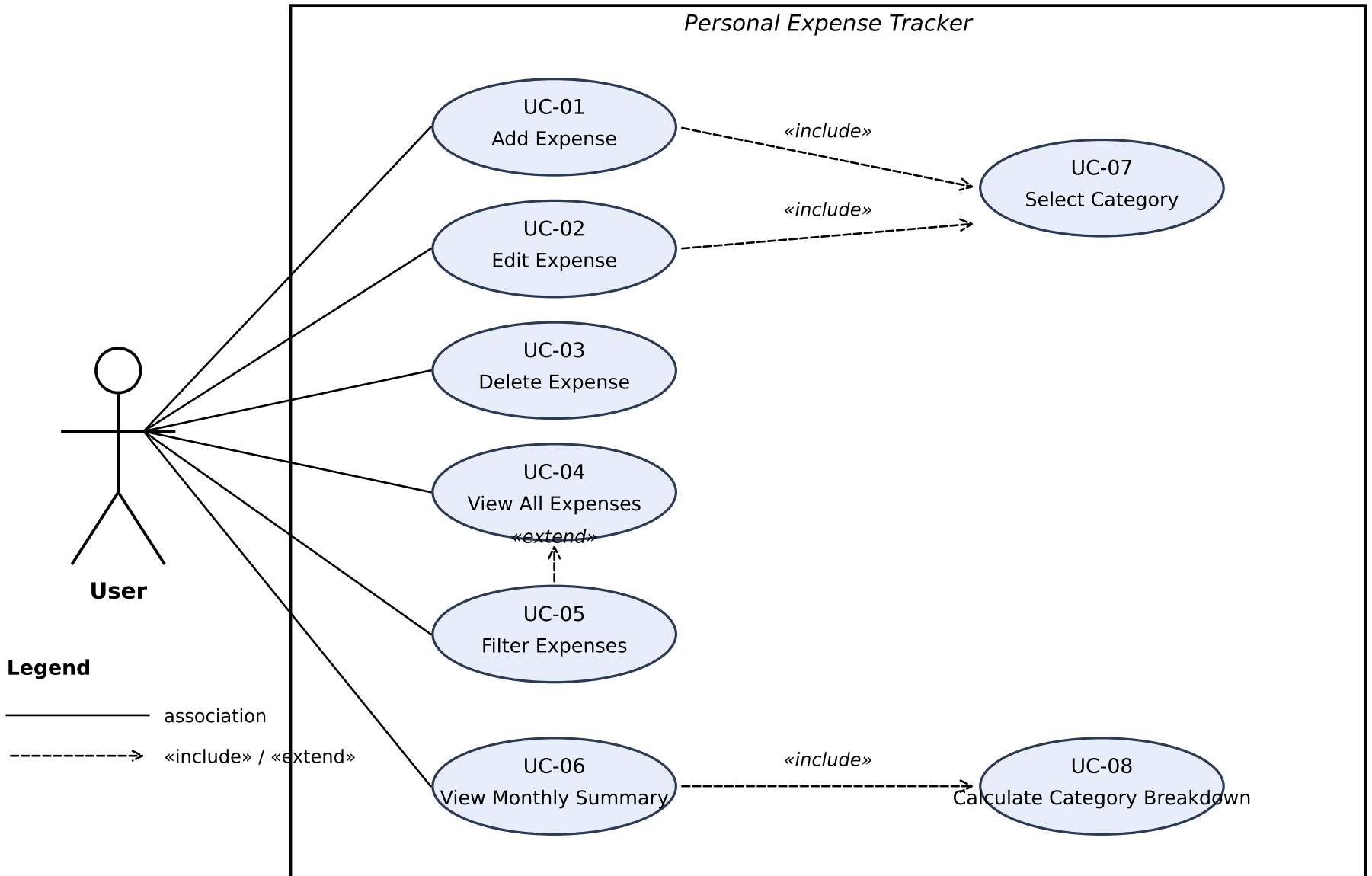


Personal Expense Tracker — Use-Case Diagram



Personal Expense Tracker — Use-Case Specification Table

ID	Name	Actor	Description	Preconditions	Main Flow	Postconditions
UC-01	Add Expense	User	Record a new expense with amount, date, description, category.	Application is running; categories list available.	1. User opens 'Add Expense' form. 2. User selects a category (UC-07). 3. User enters amount, date, description. 4. System validates input (non-null, amount ≥ 0). 5. System stores the expense.	Expense is saved and visible in the expense list.
UC-02	Edit Expense	User	Modify the fields of an existing expense.	At least one expense exists; user has selected one.	1. User selects expense and chooses 'Edit'. 2. Form pre-fills with current values. 3. User changes fields; may reselect category (UC-07). 4. System validates and replaces the stored expense.	Selected expense is updated in place.
UC-03	Delete Expense	User	Remove an existing expense from the tracker.	At least one expense exists; user has selected one.	1. User selects expense and chooses 'Delete'. 2. System removes the expense from storage. 3. List view refreshes.	Expense is no longer present.
UC-04	View All Expenses	User	Display every recorded expense in a list.	Application is running.	1. User opens the main expenses view. 2. System fetches all expenses. 3. System renders them in chronological / insertion order.	User sees the full expense list (empty list if none).
UC-05	Filter Expenses	User	Narrow the expense list by category or date. Extends UC-04.	User is on the 'View All Expenses' screen.	1. User selects a filter (category or date). 2. System computes the matching subset. 3. List view shows only the filtered results.	Only matching expenses are displayed; clearing the filter restores UC-04 view.
UC-06	View Monthly Summary	User	See total spending grouped by category for a given month/year.	Application is running; one or more expenses exist.	1. User selects month and year. 2. System runs UC-08 to compute totals per category. 3. System displays the breakdown.	Per-category totals for the selected month are shown.
UC-07	Select Category	User	Choose a category for an expense being added or edited. Included by UC-01 and UC-02.	Add or Edit Expense flow is in progress.	1. System presents available categories. 2. User picks one. 3. System validates the choice and returns control to the calling use case.	A valid category is attached to the expense in flight.
UC-08	Calculate Category Breakdown	System	Aggregate expenses by category over a month/year. Included by UC-06.	Triggered from UC-06 with a month and year.	1. System filters expenses to the given month/year. 2. System groups them by category name. 3. System sums amounts per category and returns the map.	A category → total-spend map is returned to UC-06.